

AMGA, Russian Federation

WMO: 249620

Lat: 60.900N

Lon: 131.983E

Elev: 479

StdP: 14.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-61.6	-58.5	-64.1	0.1	-61.0	-61.0	0.1	-58.2	11.6	-22.1	10.2	-22.1	0.4	210	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	86.7	66.7	82.9	64.9	79.0	63.5	69.8	80.9	67.7	78.6	65.5	75.5	6.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.1	74.3	63.9	90.5	71.9	61.6	83.5	70.2	34.2	81.4	32.5	78.8	30.8	75.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	13.7	11.7	-63.7	91.6	4.5	3.9	-67.0	94.4	-69.6	96.7	-72.2	98.9	-75.5	101.8		
			-63.4	73.3	3.9	2.8	-66.2	75.4	-68.5	77.0	-70.7	78.6	-73.5	80.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	-39.3	-33.0	-6.2	23.2	45.4	59.7	65.5	58.9	42.8	18.1	-17.9	-38.7	
	DBStd	40.14	11.94	11.29	13.86	11.19	7.23	6.94	6.49	6.76	7.79	12.57	16.73	12.68	
	HDD50	13850	2768	2323	1743	804	180	8	0	9	240	989	2036	2750	
	HDD65	18337	3233	2743	2208	1254	609	183	74	211	667	1454	2486	3215	
	CDD50	1126	0	0	0	0	36	301	481	286	22	0	0	0	
	CDD65	137	0	0	0	0	0	26	89	22	0	0	0	0	
	CDH74	2138	0	0	0	0	15	564	1212	345	2	0	0	0	
	CDH80	664	0	0	0	0	1	145	435	83	0	0	0	0	
Wind	WSAvg	4.0	1.8	1.8	3.3	5.2	6.4	5.5	4.8	4.6	4.8	4.5	3.0	1.9	
Precipitation	PrecAvg	11.1	0.4	0.3	0.2	0.3	0.9	1.5	1.9	2.2	1.6	0.9	0.7	0.4	
	PrecMax	15.9	0.8	0.7	0.8	1.1	2.8	3.6	4.8	7.9	7.2	2.2	1.7	0.7	
	PrecMin	7.6	0.1	0.1	0.0	0.0	0.2	0.4	0.5	0.0	0.3	0.2	0.2	0.1	
	PrecStd	2.6	0.2	0.2	0.2	0.3	0.6	0.7	1.1	1.7	1.3	0.5	0.4	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	8.1	37.3	57.1	76.0	88.1	92.9	87.3	71.5	45.5	29.0	5.7	
		MCWB	0.7	6.8	31.7	43.1	56.7	64.8	69.5	67.9	57.0	40.0	27.8	5.4	
	2%	DB	-9.4	-2.9	30.2	50.2	69.8	84.2	88.7	81.7	64.7	39.8	23.1	-7.7	
		MCWB	-9.9	-3.4	25.9	38.9	52.8	64.2	68.1	65.5	54.1	35.6	22.3	-8.2	
	5%	DB	-15.8	-10.1	23.7	45.0	65.3	80.6	85.1	77.6	60.4	36.2	14.7	-14.1	
		MCWB	-16.2	-10.6	20.6	36.2	50.3	62.3	66.7	63.1	52.0	33.0	13.9	-14.5	
	10%	DB	-21.1	-15.4	17.6	40.6	61.0	76.7	81.2	73.8	56.2	33.2	6.8	-20.3	
		MCWB	-21.3	-15.8	15.4	33.8	48.0	60.9	65.5	61.4	49.5	31.2	6.2	-20.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	7.1	31.4	43.4	58.3	69.3	73.4	70.8	58.5	40.9	28.0	5.4
			MCDB	1.6	8.0	37.1	54.6	73.1	82.3	84.9	79.3	67.2	45.1	28.7	5.8
2%		WB	-9.8	-3.4	26.0	39.7	54.1	66.2	71.0	67.9	56.0	36.2	22.3	-8.2	
		MCDB	-9.5	-2.9	30.2	48.9	67.0	78.4	82.9	77.8	62.3	39.1	23.0	-7.9	
5%		WB	-16.1	-10.6	20.7	36.8	51.2	64.2	69.0	65.4	53.2	33.6	14.1	-14.5	
		MCDB	-15.8	-10.1	23.8	43.9	62.3	76.0	80.6	74.9	58.5	35.6	14.7	-14.2	
10%		WB	-21.4	-15.8	15.3	34.2	48.9	62.2	67.2	62.8	50.4	31.1	6.2	-20.5	
		MCDB	-21.2	-15.5	17.7	40.0	59.0	74.2	78.0	71.2	55.3	33.1	6.8	-20.2	
Mean Daily Temperature Range		5% DB	MDBR	13.0	19.2	31.2	27.0	21.7	25.0	23.1	22.4	18.7	14.7	14.8	11.7
			MCDBR	20.0	24.2	33.6	26.4	31.5	33.5	32.0	30.2	26.8	15.9	17.9	18.9
	MCWBR		19.9	23.7	29.3	16.7	16.7	15.7	14.0	14.9	16.2	11.5	17.4	18.8	
	5% WB	MCDBR	20.1	24.0	33.2	24.8	27.0	28.6	25.8	25.5	22.4	14.0	18.0	19.0	
		MCWBR	19.8	23.5	29.1	16.3	14.8	14.4	12.8	13.7	14.3	10.6	17.5	18.8	
Clear-Sky Solar Irradiance	taub	0.223	0.265	0.306	0.371	0.466	0.419	0.507	0.448	0.353	0.295	0.232	0.193		
	taud	1.993	2.065	2.051	2.008	1.986	2.257	1.983	2.127	2.369	2.275	2.006	1.906		
	Ebn at Noon	160	224	253	256	239	257	226	231	240	214	150	116		
	Edh at Noon	14	24	35	45	50	39	50	40	26	20	13	9		
All-Sky Solar Radiation	RadAvg	144	476	1012	1586	1659	1872	1720	1335	837	398	191	75		
	RadStd	8	19	38	57	145	118	104	102	69	42	16	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.15	+3.71	+2.84	N/A	N/A	N/A	-362	-320	N/A	N/A	
(47)	Regional (6 neighbors)	+1.17	+2.14	+1.89	N/A	N/A	N/A	-396	-418	N/A	N/A	

Nomenclature: See separate page